

Alan (Qi) Xu

Engineering Leader · Staff Software Engineer

Toronto, Canada · +1 416 835 5956 · alanxu.80@gmail.com · LinkedIn

SUMMARY

- **Tech Lead for AI automation at Meta** — architect autonomous, self-improving agents and the RAG, evaluation, and governance platforms, boosting the dev efficiency of thousands of Meta engineers.
- **Lead architect of the Analytical Platform at CPP Investments** — the firm-wide research and elastic compute platform that put distributed computing in the hands of every Public Capital Markets team.
- **20+ years building large-scale distributed systems** across finance, telecom, and e-commerce — microservices, event-driven architecture, and ultra-low-latency messaging at organizations including CPP Investments, Canonical (Ubuntu), RSA, Thomson Reuters, and Bank of China.
- **Proven technical leadership**: set multi-year platform direction across partner orgs at Meta, led architecture for CPP Investments' Analytical Platform, and managed client-facing delivery teams and project portfolios at Canonical.
- **Core strengths**: AI agent systems & RAG · distributed systems architecture · platform & developer productivity · technical leadership and mentorship · cross-org program ownership

EXPERIENCE

Meta — Tech Lead / Staff Engineer

Jan 2022 – Present

- **Lead AI automation programs** across Meta's Monetization org, architecting autonomous, self-improving agents that measurably elevated the development efficiency of thousands of Meta engineers.
- **Built reliable model training infrastructure** that enabled Meta's online and recurring model training, supporting continuous model improvement and faster iteration across Monetization AI systems.
- **Architected and evolved a production RAG system** that dramatically improved agent context intelligence, tuning retrieval and ranking to reach an optimal recall/precision balance for code and knowledge workloads.
- **Drove the development of evaluation, tracing, governance, and agent-collaboration platforms**, lifting agent reliability, task completion rate, and end-to-end latency — turning agent quality from anecdote into a measurable, regression-tested property.
- **Set technical direction and grew the team** — defined the agent architecture and platform roadmap, aligned partner orgs on shared standards, drove adoption beyond the founding team, and mentored senior and mid-level engineers on agent design and evaluation methodology.

CPP Investments (CPPIB) — Lead Engineer & Architect

Dec 2017 – Jan 2022

- **Lead architect for the Analytical Platform** — designed highly scalable, available services and an elastic Jupyter/Kubernetes compute layer that put distributed computing in the hands of non-technical quantitative researchers, materially improving the efficiency and ROI of every Public Capital Markets team.
- **Led the engineering team** — set technical direction, owned design and code review standards, mentored developers, and partnered directly with business stakeholders to translate research needs into a delivery roadmap.
- **Designed a configuration-driven data ingestion pipeline** on AWS (Airflow, EMR) that scaled to the firm's growing data footprint without per-source engineering work.
- **Led the migration from legacy monoliths to microservices** — redesigned domain boundaries, built the CI/CD pipelines behind independent deployment, and introduced event-driven infrastructure that lifted throughput and availability.

Canonical (Ubuntu) — Cloud Engagement Manager

Apr 2014 – Apr 2017

- **Owned professional-services engagements end-to-end** — architecture, scoping, delivery, and customer relationship — for major telecom customers adopting OpenStack and microservice architectures.
- **Managed delivery teams and project portfolios** from design through go-live, balancing customer commitments against engineering capacity.
- Built and maintained SaaS/microservice applications supporting the Ubuntu ecosystem.

EMC — Practice Consultant

Sep 2012 — Apr 2014

- **Designed and delivered a cloud-based fraud protection service** for Bank of China's online banking — a high-performance web service serving a billion-user population with rich monitoring and reporting.
- **Owned performance engineering** for a mission-critical system, meeting hard SLAs of >5,000 TPS and <200 ms average response time.
- Ran the full delivery lifecycle: requirements, design, development, functional and performance testing, deployment.

Thomson Reuters — Senior Software Engineer

Nov 2007 — Sep 2012

- **Lead engineer for Trade Notification Service**, a low-latency, high-availability FX trade messaging system (FIX / FpML) used by 90% of the world's top financial institutions.
- Coordinated design and delivery across engineering teams on three continents in an agile environment.

Synnex China — Senior Software Engineer / Architect

Nov 2005 — Oct 2007

- **Architect and lead engineer of Synnex Product Search Engine**, a standalone high-performance search and indexing service powering the company's B2B e-commerce and supply chain products.
- Lead engineer for EC Express, a large-scale J2EE B2B e-commerce platform for Synnex's global IT distribution business.

Datang Telecom — Software Engineer

Sep 2004 — Jun 2005

- Delivered and supported the Remote Personalize System, a SIM provisioning platform deployed across most China Mobile branches; the project won the Ministry of Information Industry's top innovation award.

TECHNICAL SKILLS

AI / ML Systems — LLM agent architecture, autonomous & self-improving agents, RAG (retrieval, ranking, chunking, embeddings), agent evaluation & benchmarking, tracing/observability, model governance, prompt and context engineering; agent frameworks (LangChain, Llamaindex, LangGraph, AutoGen, Semantic Kernel, DSPy, CrewAI)

Languages — Python, Java, JavaScript/TypeScript, SQL, Shell

Architecture — Microservices, event-driven systems, domain-driven design, distributed systems, low-latency messaging, design patterns

Cloud & Infrastructure — AWS (EMR, Athena, Glue, Step Functions, EKS/ECS, S3, DynamoDB, Redshift, API Gateway, ELB, SNS/SQS, IAM), GCP, Kubernetes, Docker, Terraform, Helm, OpenStack (Juju, MAAS), CI/CD

Cloud AI Infrastructure — AWS Bedrock & SageMaker, GCP Vertex AI, model serving & inference infrastructure, GPU/accelerator compute and scheduling, vector databases (pgvector, FAISS), feature and embedding pipelines, MLOps/LLMOps tooling

Data — Spark, Airflow, Kafka, Pandas, Jupyter, Oracle, MySQL, SQL Server, MongoDB, Redis, ELK

Backend / Frontend — Spring Boot / MVC / Data / Security, Spring Cloud (Eureka, Ribbon, Hystrix, Feign, Zuul), Java EE, JPA, Flask, Angular, Node.js, REST APIs

Leadership — Technical strategy & roadmapping, cross-org program leadership, mentorship & career development, agile/scrum, stakeholder and customer management

EDUCATION & CERTIFICATIONS

Master of Software Engineering — Beihang University (BUAA), Beijing

AWS Certified Solutions Architect – Associate

Project Management Professional (PMP)

SELECTED CLIENTS & PARTNERS

China Mobile · Deutsche Telekom · Bank of China · Credit Suisse · NYSE · ICAP · CPP Investments · Huawei · Quanta · British Sky Broadcasting · MTRC